2024 National College Entrance Examination

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I. Single-Choice Questions

1. If $z = \sqrt{2}i$, then $z\overline{z} = (\ )$

(A) -2 (B) $-\sqrt{2}$ (C) $\sqrt{2}$ (D) 2

2. Given the sets $A = \{1, 2, 3, 4, 5, 9\}$ and $B = \{x \mid x + 1 \in A\}$, then $A \cap B = (\ )$

(A) $\{1, 2, 3\}$ (B) $\{3, 4, 9\}$

(C) $\{1, 2, 3, 4\}$ (D) $\{2, 3, 4, 5\}$

3. Let vectors $\boldsymbol{a} = (x + 1, x)$, $\boldsymbol{b} = (x, 2)$, $\boldsymbol{c} = (1, 1)$. If $\boldsymbol{a} - 2\boldsymbol{b}$ is collinear with $\boldsymbol{c}$, then $x = (\ )$

(A) $\frac{5}{2}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{5}{2}$

4. In the final round of a solo singing competition, four participants—A, B, C, and D—each perform once, and the performance order is determined by random draw. What is the probability that C is not the first to perform and either A or B is the last to perform?

(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

5. Let S_n denote the sum of the first n terms of the arithmetic sequence $\{a_n\}$. Given that $S_9 = 1$, then $a_3 + a_7 = (\ )$

(A) $\frac{1}{9}$ (B) $\frac{2}{9}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

6. The two foci of a hyperbola are at $(0, 4)$ and $(0, -4)$, and the point $(-6, 4)$ lies on the hyperbola. What is the eccentricity of the hyperbola?

(A) 4 (B) 3 (C) 2 (D) $\sqrt{2}$

7. Let the function $f(x) = \frac{e^x + 2\sin x}{1 + x^2}$. Then the area of the triangle formed by the tangent to the curve $y = f(x)$ at the point $(0, 1)$ and the two coordinate axes is $(\ )$

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

8. The graph of the function $y = -x^2 + (e^x - e^{-x})\sin x$ over the interval $[-2.8, 2.8]$ is approximately ( )

(A) *[Option A]* (B) *[Option B]*

(C) *[Option C]* (D) *[Option D]*

9. The line $2x - y - 2 = 0$ intersects the circle $x^2 + y^2 - 6x - 8y = 0$ at points A and B. Then $|AB| =$ ( )

(A) 4 (B) 5 (C) 8 (D) 10

10. Given $\frac{\cos\alpha}{\cos\alpha - \sin\alpha} = \sqrt{3}$, then $\tan\left(\alpha + \frac{\pi}{4}\right) =$ ( )

(A) $2\sqrt{3} + 1$ (B) $2\sqrt{3} - 1$ (C) $\frac{\sqrt{3}}{2}$ (D) $1 - \sqrt{3}$

11. Let α, β be two planes, m, n be two lines, and $\alpha \cap \beta = m$. The following four propositions:

① If $m \parallel n$, then $n \parallel \alpha$ or $n \parallel \beta$;
② If $m \perp n$, then $n \perp \alpha$ or $n \perp \beta$;
③ If $n \parallel \alpha$ and $n \parallel \beta$, then $m \parallel n$;
④ If the angle between n and α, β are equal, then $m \perp n$.

The numbers of all true propositions are( )

(A) ①③ (B) ②④ (C) ①②③ (D) ①③④

12. Let the sides opposite angles A, B, C of $\triangle ABC$ be a, b, c respectively. Given $B = 60^\circ$, $b^2 = \frac{9}{4}ac$, then $\sin A + \sin C =$ ( )

(A) $\frac{3}{2}$ (B) $\sqrt{2}$ (C) $\frac{\sqrt{7}}{2}$ (D) $\frac{\sqrt{3}}{2}$

II. Fill-in-the-Blank Questions

13. The maximum value of the function $f(x) = \sin x - \sqrt{3}\cos x$ on the interval $[0, \pi]$ is _____.

14. Given that the upper and lower base radii of frustums A and B are both r_1 and r_2 respectively, and the slant heights of frustums A and B are $2(r_2 - r_1)$ and $3(r_2 - r_1)$ respectively, the ratio of the volumes of frustum A to frustum B is _____.

15. Given $a > 1$ and $\frac{1}{\log_8 a} - \frac{1}{\log_a 4} = -\frac{5}{2}$, then $a =$_____.

16. When $x > 0$, the curves $y = x^3 - 3x$ and $y = -(x-1)^2 + a$ have two intersection points; then the range of values for a is _____.

III. Free-Response Questions

17. Let S_n be the sum of the first n terms of the geometric sequence $\{a_n\}$, given $2S_n = 3a_{n+1} - 3$.

(1) Find the general term of $\{a_n\}$;

(2) Find the sum of the first n terms of the sequence $\{S_n\}$.

18. A factory is upgrading its production line with intelligent transformation. After the upgrade, 150 products are randomly sampled from workshops A and B for inspection, with the following data:

	top-grade	qualified	defective	total
workshop A	26	24	0	50
workshop B	70	28	2	100
total	96	52	2	150

(1) Fill in the following contingency table:

	top-grade	non-top-grade
workshop A		
workshop B		

Can we be 95% confident that there is a difference in the top-grade rates between workshops A and B? Can we be 99% confident that there is a difference in the top-grade rates between workshops A and B?

(2) It is known that before the upgrade, the top-grade rate of the factory's products was $p = 0.5$. Let $\overline{p}$ be the top-grade rate of the n sampled products after the

upgrade. If $\overline{p} > p + 1.65\sqrt{\frac{p(1-p)}{n}}$, then the top-grade rate is considered to have increased. Based on the data from the 150 sampled products, can we conclude that the top-grade rate of the factory's products has increased after intelligent transformation? ($\sqrt{150} \approx 12.247$)

Reference: $K^2 = \frac{n(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$

$\frac{P(K^2 \geq k)}{k}$|0.050 0.010 0.001

3.841 6.635 10.828

19. As shown in the figure, in a pentahedron with vertices A, B, C, D, E, F, quadrilaterals $ABCD$ and $ADEF$ are both isosceles trapezoids, $EF \parallel AD$, $BC \parallel AD$, $AD = 4$, $AB = BC = EF = 2$, $ED = \sqrt{10}$, $FB = 2\sqrt{3}$, and M is the midpoint of AD.

[Figure]

(1) Prove that $BM \parallel$ plane CDE;

(2) Find the distance from M to plane FAB.

20. Given the function $f(x) = a(x-1) - \ln x + 1$.

(1) Discuss the monotonicity of $f(x)$;

(2) Let $a \leqslant 2$, prove that when $x > 1$, $f(x) < \mathrm{e}^{x-1}$.

21. Let the ellipse $C : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1(a > b > 0)$ have its right focus at F, and the point $M\left(1, \frac{3}{2}\right)$ lies on C, with $MF \perp x - axis$.

(1) Find the equation of C;

(2) A line passing through point $P(4, 0)$ intersects C at points A and B. Let N be the midpoint of segment FP. The line NB intersects line MF at point Q. Prove that $AQ \perp y - axis$.

22. In the rectangular coordinate system xOy, with the origin as the pole and the positive x-axis as the polar axis, establish a polar coordinate system. The polar equation of curve C is $\rho = \rho\cos\theta + 1$.

(1) Write the Cartesian equation of C;

(2) Let line l :

$$\begin{cases} x = t, \\ y = t + a \end{cases}$$

(where t is a parameter). If C intersects l at points A and B, and $|AB| = 2$, find a.

23. Given real numbers a, b satisfying $a + b \geqslant 3$.

(1) Prove that $2a^2 + 2b^2 > a + b$;

(2) Prove that $|a - 2b^2| + |b - 2a^2| \geqslant 6$